Chapter 2 Autonomous Systems and Intelligent Agents

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2.1 Definitions and Scopes

Autonomous systems and intelligent agents are two central concepts in the evolving landscape of automation and artificial intelligence (AI). Both play pivotal roles in driving innovation across various industries, from transportation to healthcare. This chapter provides an overview of these concepts, exploring their significance and influence in shaping modern technological advancements.

2.1.1 What Are Autonomous Systems?

Autonomous systems are machines or technologies designed to operate independently without continuous human oversight. These systems autonomously perform tasks by leveraging sensors to gather data, advanced algorithms to process and interpret this information, and actuators to execute necessary actions toward achieving predefined goals. The range of autonomous systems varies widely, from simple devices such as thermostats, which adjust the temperature based on environmental inputs, to complex technologies such as self-driving cars and unmanned aerial vehicles that navigate dynamic environments with adaptive decision-making.

At their core, autonomous systems follow a three-step process:

1. Sensing: The system uses sensors, such as cameras, light detection and ranging (LiDAR), or thermometers to gather data from its surroundings.
2. Decision-making: Algorithms, often powered by AI, process the sensor data to make informed decisions about how to react to the environment.
3. Acting: The system then executes actions, such as steering a vehicle, adjusting settings, or communicating with other systems.

These systems can be physical entities, such as robots, or software-based, such as automated trading platforms or AI-driven recommendation engines. The defining feature of autonomous systems is their capacity to operate independently, making decisions and taking actions without human input.

2.1.2 What Are Intelligent Agents?

An intelligent agent is a system that perceives its environment, makes decisions, and takes actions to achieve goals while learning and adapting from interactions. Unlike basic autonomous systems, intelligent agents use AI algorithms to process data and respond to changing conditions. Examples include virtual assistants such as Siri or Alexa, which perform tasks and learn from user interactions. In more complex applications, in-

telligent agents manage large-scale systems such as smart grids, adjusting energy distribution in real time for efficiency.

Autonomous systems operate independently but follow preprogrammed rules without learning or adaptation, such as a robot vacuum that navigates and cleans based on preset instructions. In contrast, intelligent agents can learn, adapt, and make proactive decisions, such as a self-driving car that continuously learns from its environment, predicts traffic patterns, and communicates with other systems to enhance safety and efficiency.

2.1.3 Degrees of Autonomy: From Manual to Full Autonomy

The levels of autonomy in autonomous systems and intelligent agents define the degree of independence and intelligence these systems exhibit when perceiving their environment, making decisions, and executing actions. These levels range from full human control to complete system autonomy, showing how much responsibility shifts from humans to machines.

1. Manual Control (No Autonomy): The system operates entirely under human control with no independent decision-making. Early examples include basic robots and remotely controlled drones, where every action is directed by a human.
2. Assisted Control (Level 1 Autonomy): The system automates specific tasks but decision-making remains human-driven. For instance, adaptive cruise control in cars adjusts speed, but the driver controls steering and other critical functions.
3. Partial Automation (Level 2 Autonomy): The system can control multiple functions independently, such as speed and steering, but requires human oversight. Semi-AVs and autopilot systems in aircraft fall into this category.
4. Conditional Automation (Level 3 Autonomy): The system manages most functions but requires human intervention in complex situations. An example is an autonomous car that drives on highways but hands control to the human driver in urban settings.

5. High Automation (Level 4 Autonomy): The system handles nearly all operations independently, even in complex environments. Human intervention is possible but rarely necessary, except in extreme conditions such as severe weather.
6. Full Automation (Level 5 Autonomy): The system operates entirely without human involvement, capable of handling any task and environment autonomously. Fully self-driving vehicles represent this ultimate level of autonomy, requiring no manual controls or human oversight.

Achieving full autonomy in all applications remains a significant challenge due to the complexity of real-world environments and the limitations of current technologies. However, the journey toward full autonomy continues to drive research and development in AI, machine learning (ML), and robotics.

2.1.4 Overview of Applications and Impact on Various Industries

Autonomous systems and intelligent agents are having a profound impact across numerous industries by fostering innovation, improving efficiency, and unlocking new opportunities. Their diverse applications offer distinct advantages while also presenting specific challenges.

In transportation, self-driving cars and autonomous vehicles (AVs) are revolutionizing personal transport and logistics, while drones are reshaping delivery systems (see Gangwani and Gangwani 2021). Healthcare benefits from surgical robots and AI-powered diagnostics, improving patient care and elderly assistance (see Guni et al. 2024). In manufacturing, industrial robots boost productivity, while intelligent agents optimize supply chains and enable smart factories (see Peres et al. 2020). Agriculture is seeing increased efficiency with autonomous tractors, drones, and smart farming practices that optimize irrigation and pest control (see Misra et al. 2022). In finance, automated trading systems, chatbots, and AI-driven algorithms are reshaping financial services, enhancing speed, accuracy, and fraud detection (see Dakalbab et al. 2024). The entertainment industry uses intelligent agents for personalized content, AI-generated media,

and dynamic gaming experiences (see Anantrasirichai and Bull 2022). In military and defense, AVs and drones perform reconnaissance and combat missions, reducing risks to human soldiers but raising ethical concerns (see Raska and Bitzinger 2023; Rickli and Mantellassi 2023; Soorchaei et al. 2022). As these technologies evolve, they will increasingly shape the future, driving growth and reshaping how we live and work.

2.2 Core Concepts and Components

Autonomous systems have rapidly evolved from experimental concepts into transformative technologies that affect numerous industries. These systems are designed to perform tasks without continuous human guidance, utilizing advanced algorithms and sensing technologies to interact with the environment and make intelligent decisions. In this chapter, we will explore the core technologies and components that make autonomy possible. Particular attention will be paid to AI and ML, which are critical to the decision-making and perception functions of autonomous systems. We will also dive deeply into perception, decision-making, and action, the heart of autonomous operation, and discuss the role of communication between agents and external entities.

2.2.1 Artificial Intelligence and Machine Learning

At the core of any advanced autonomous system lies AI and ML. These technologies form the foundation for a system's ability to interpret the world, make decisions, and learn from interactions.

2.2.1.1 Basics of AI and ML

AI refers to the science of creating machines or systems that can perform tasks that require human intelligence. It includes techniques such as natural language processing, image recognition, and decision-making. AI encompasses a broad range of algorithms, from rule-based systems (simple, logical reasoning systems) to more sophisticated ML algorithms that can learn patterns from data without explicit programming. ML, a subset of AI, focuses on developing systems that learn from experience. Traditional programming involves manually coding every instruction, but ML allows systems to learn by example. Based on data, the system identifies patterns

and makes predictions or decisions based on new information. The three main categories of ML problems are:

1. Supervised Learning: In this approach, the system is trained on a dataset that contains input–output pairs. The system learns to map the input to the correct output, which can then be generalized to new, unseen data.
2. Unsupervised Learning: This method uses unlabeled data. The system is tasked with identifying patterns or clusters in the data without prior knowledge of the correct output.
3. Reinforcement Learning (RL): In this approach, the system learns by interacting with an environment and receiving feedback in the form of rewards or punishments. The goal is to maximize long-term rewards by making a series of decisions (Wiering and Van Otterlo 2012).

2.2.1.2 Role in Autonomous Systems

In autonomous systems, AI and ML are essential for enabling machines to perform tasks without human oversight. These technologies allow systems to:

- Perceive Their Environment: ML algorithms help autonomous systems process data from sensors (such as cameras and LiDAR) to understand objects, paths, and events around them.
- Make Decisions: AI-based decision-making processes enable the system to choose actions that align with its goals, whether navigating a vehicle, controlling a robotic arm, or optimizing resource usage in a smart grid.
- Adapt and Learn: By learning from past experiences, autonomous systems can improve their performance over time, becoming more efficient and capable of handling more complex tasks.

AI and ML are particularly useful in applications where systems operate in dynamic environments, such as autonomous drones or self-driving cars. By processing sensor data in real time, these systems can identify changes in their surroundings and adjust their behavior accordingly.

2.2.2 Perception, Decision-making, and Action

The central feature of any autonomous system is its ability to sense the environment, make decisions based on that information, and take appropriate action. These three processes, i.e. perception, decision-making, and action, are tightly intertwined and enable systems to operate autonomously. In this section, we will delve deeper into these essential processes.

2.2.2.1 Perception

Perception can have different meanings depending on the context. In general, perception involves collecting data about the environment using sensors and interpreting that data to form a coherent understanding of the environment. In the domain of robotics, perception refers to an autonomous agent's ability to be aware of its own state and surroundings, in order to make and execute informed decisions for a planned time horizon. In this context, perception involves sensing the dynamic environment around the agent, building a reliable and detailed representation of this environment, and understanding the relative relationship between the agent and its surroundings based on the sensory data (Liu et al. 2018). The core components of a perception unit are listed as follows. It is worth mentioning that autonomous systems in other domains include some (but not all) of these components in their perception unit.

Sensing is a multistep process that begins with capturing diverse sets of raw environmental data using hardware devices such as cameras, LiDAR, radio detection and ranging (radar), ultrasonic sensors, GPS, and inertial measurement units (IMUs). Each sensor provides specific information, such as visual images, depth perception, speed, orientation, and spatial distance. Since no single sensor can offer a complete view, data fusion merges inputs from multiple sensors, compensating for individual limitations and producing a more accurate, holistic understanding of the environment. Before further analysis, the data undergoes preprocessing, where it is cleaned and structured. Techniques such as noise reduction, normalization, and distortion correction eliminate inaccuracies, ensuring the data is optimized for computational tasks. This integration of sensor

technology, real-time processing, and refinement allows for a precise and detailed interpretation of the environment.

Scene understanding, localization, and mapping is a comprehensive process combining object recognition, scene interpretation, positioning, and map creation to support effective navigation and decision-making. It starts with object detection and classification, where ML techniques, such as convolutional neural networks (CNNs), identify and categorize key elements such as pedestrians, vehicles, obstacles, and road signs. Beyond detection, scene understanding analyzes relationships between objects, interprets dynamic situations such as traffic flow, and assesses potential hazards using techniques such as scene segmentation and contextual reasoning.

Localization ensures the system knows its exact position and orientation within the environment. Methods such as simultaneous localization and mapping (SLAM), GPS integration, and visual odometry maintain accurate, real-time location tracking. The final step is mapping, where the system builds and updates a dynamic representation of its surroundings using techniques such as three-dimensional (3D) mapping and occupancy grids. This continuously updated map informs path planning and obstacle avoidance. By integrating these elements, the system achieves a high level of situational awareness and precision, enabling it to navigate and operate safely in complex real-world environments.

2.2.2.2 Decision-making Processes, Planning, and Navigation

Decision-making in autonomous systems and intelligent agents involves selecting actions or a set of actions based on environmental perception and predefined goals. The capacity to autonomously make decisions is essential for these systems to operate without human intervention, particularly in dynamic and complex environments. This process involves several techniques, including rule-based algorithms, learning-based models, and optimization methods.

- Rule-based Decision-making: This is a fundamental approach where systems follow predefined rules or logic. For instance, an AV programmed to stop upon detecting an obstacle is an example of this

method. While rule-based decision-making is straightforward to implement, it can struggle with highly variable or unpredictable environments due to its lack of adaptability.

- Learning-based Decision-making: Learning-based approaches, such as those powered by AI and ML, allow systems to improve over time based on historical data or direct experience. For instance, RL enables agents to interact with their environment and learn the optimal actions by receiving feedback in the form of rewards or penalties. A robotic vacuum learning a room's layout through trial and error to optimize its cleaning path exemplifies this method.
- Optimization and Probabilistic Methods: These techniques involve algorithms that search for the best course of action from multiple possibilities, often accounting for uncertainty in the environment. For example, path planning algorithms used by autonomous drones calculate the shortest or safest route. Probabilistic methods, such as Bayesian networks, help make decisions in environments where sensor data may be incomplete or noisy.

Planning is the process by which an autonomous system determines the sequence of actions needed to achieve a goal. This involves considering the system's objectives, constraints (e.g. physical limitations or energy), and possible future states. Path planning algorithms, such as A* or Dijkstra's algorithm, are commonly used to find the most efficient route from point A to point B. More advanced systems may use probabilistic or optimization-based planning methods to account for uncertainties and dynamic environments.

Navigation refers to the system's ability to move through the environment safely and efficiently while following the planned route. Autonomous systems use sensor data and localization techniques (e.g. GPS, SLAM) to understand their position and surroundings. These systems continuously update their trajectory based on real-time inputs, ensuring they avoid obstacles and stay on course. Together, decision-making, planning, and navigation work together to enable autonomous systems and intelligent agents to navigate complex environments, complete tasks, and make real-time decisions with minimal human input.

2.2.2.3 Actuation and Interaction with the Environment

Actuation and control are essential components of autonomous systems, converting high-level decisions into physical actions that enable interaction with the environment and task execution. These systems aim to operate smoothly, accurately, and safely while navigating space, manipulating objects, or adapting to complex environments. Actuation refers to the physical mechanisms driving movement and actions. For example, in AVs, actuators manage steering, acceleration, braking, and other mechanical functions, while in robotic arms, they control joint movements and tool manipulation. Key types of actuators include:

1. Electric Motors: Widely used in vehicles, drones, and robotic arms, they provide precise control over both rotational and linear movements.
2. Hydraulic and Pneumatic Actuators: Used in systems requiring high force, such as industrial robots, hydraulic actuators use fluid pressure, while pneumatic actuators rely on compressed air.
3. Linear Actuators: Convert rotational motion into straight-line movement, ideal for robotic grippers or sliders.
4. Piezoelectric Actuators: Employed in applications needing fine movements, such as medical devices or precision manufacturing, known for their accuracy.

Control in autonomous systems involves the algorithms and methods that manage the actuators to achieve desired outcomes. The control system interprets high-level commands (such as a desired speed or trajectory) and translates them into specific actions that the actuators can execute. The goals of control include maintaining stability, ensuring accuracy, and providing a responsive system that can adapt to changes in real time. Listed below are the primary control methods used:

1. Proportional Integral Derivative (PID) Control: PID control is a fundamental method widely used in many autonomous systems due to its simplicity and effectiveness. It continuously calculates the error between a desired setpoint (e.g. a specific speed or position) and the current state and then applies corrections based on three components: proportional, integral, and derivative. Proportional control corrects

the error based on its current size; integral control addresses accumulated past errors; and derivative control anticipates future errors based on the current rate of change. This method is particularly effective for systems such as speed regulation in vehicles or position control in robotics.

2. Model Predictive Control (MPC): MPC is an advanced control method that uses a mathematical model of the system to predict its future behavior and optimize control actions over a set horizon. MPC considers the current state, possible future states, and any constraints (such as avoiding obstacles) to determine the best course of action. This predictive capability makes MPC especially useful in complex, dynamic environments, such as autonomous driving, where the system needs to anticipate and respond to changes proactively (Schwenzer et al. 2021).
3. Adaptive Control: Adaptive control adjusts the control parameters in real time in response to changes in the system or its environment. This is essential for systems operating in unpredictable conditions, as it allows the control system to maintain optimal performance even when the system dynamics or external conditions vary. Adaptive control is often used in robotics, where external forces or varying loads may affect the system's behavior.
4. Fuzzy Logic Control: It handles uncertainty and imprecision by allowing for varying degrees of truth rather than binary decisions. This type of control is useful in situations where the system operates in an environment with unclear boundaries or where human-like reasoning is required. Fuzzy logic control is often applied in consumer electronics and autonomous systems requiring smooth, adaptive responses to complex inputs.
5. RL-based Control: RL is a method where the system learns to make decisions by interacting with its environment. The system receives feedback in the form of rewards or penalties and adjusts its actions to maximize long-term rewards. This method is particularly promising for complex, nonlinear control problems, such as autonomous driving or robotic manipulation, where the environment is dynamic and uncertain. Over time, the system learns an optimal policy for control, even in challenging scenarios.

The primary goals of actuation and control in autonomous systems are precision, stability, responsiveness, efficiency, and safety. In AVs, precision ensures tasks such as lane keeping are executed accurately, while stability maintains control despite disturbances. Responsiveness allows quick adaptation to changes, essential in dynamic situations like obstacle avoidance. Efficiency minimizes energy use, crucial for systems with limited power like drones. Safety ensures the system detects hazards and operates securely. In summary, actuation and control enable autonomous systems to perform complex tasks effectively and safely, supporting applications from transportation to industrial automation.

2.2.3 Communication and Interaction

Communication is essential in autonomous systems and intelligent agents, enabling them to exchange information, collaborate, and achieve shared goals. These systems rely on communication to interact with one another, their environment, and human users, ensuring efficiency and coordinated behavior.

2.2.3.1 Agent Communication Languages and Protocols

In autonomous systems, communication refers to the transmission of data, signals, or information between agents or between agents and their environment. It can take various forms, including direct communication through wireless networks or indirect communication (stigmergy), where agents modify their environment and other agents interpret those changes. Broadcast communication, where information is shared with all agents, and negotiation-based communication are also common. To ensure seamless and reliable data exchange, autonomous systems require standardized communication protocols, particularly in multi-agent systems or connected environments:

- Vehicle-to-Everything (V2X) Communication: In autonomous driving, V2X enables data exchange between vehicle-to-vehicles (V2V), vehicle-to-infrastructure (V2I), and vehicle-to-pedestrians (V2P). This real-time communication enhances safety by updating vehicles on traffic conditions, road hazards, and pedestrian activity.

- Message Passing and Wireless Communication: In multi-agent systems, message-passing protocols such as Transport Control Protocol/Internet Protocol and User Datagram Protocol facilitate direct communication between agents. These protocols allow agents to update each other, share status information, and collaborate on tasks. Systems such as Wi-Fi, Bluetooth, and 5G are often used for communication between devices, agents, or with the cloud.
- Agent Communication Languages: Intelligent agents often communicate using specialized languages such as the Knowledge Query and Manipulation Language or the Foundation for Intelligent Physical Agents, which define rules for message structure and interaction between intelligent agents to support effective communication and ensure consistent exchanges. In robotics, Robot Operating System facilitates distributed communication between system components, enabling interoperability. These protocols enable structured communication, allowing agents to share knowledge, negotiate, or coordinate actions.
- Internet of Things Protocols: Protocols such as MQTT and CoAP are often used for low-power communication in systems with multiple autonomous devices or smart systems.

2.2.3.2 Communication's Role in Coordination and Cooperation and Challenges

A critical role of communication is enabling coordination and cooperation.

Multi-agent systems, such as AV fleets or robotic swarms, depend on communication to align actions and prevent conflicts, such as avoiding collisions or coordinating collective tasks. Without communication, these systems risk inefficiency or failure in achieving their objectives.

Despite its importance, communication in autonomous systems faces challenges. Latency and reliability are critical concerns in time-sensitive applications such as autonomous driving, where delays can have severe consequences. Scalability becomes a challenge as the number of communicating agents increases. Security is vital, as malicious agents could disrupt communication or provide false information. Finally, ensuring cor-

rect interpretation of received data is essential, especially when humans are involved.

In conclusion, communication is a foundational aspect of autonomous systems and intelligent agents. It enables coordination, cooperation, and efficient decision-making. As these systems become more prevalent, the need for robust, scalable, and secure communication protocols will grow to ensure their reliable and effective operation.

2.3 Applications and Case Study: Autonomous Vehicle

Autonomous driving stands as a transformative innovation in autonomous systems, where vehicles are designed to operate independently without human input. By leveraging advanced sensors, AI, ML, and control systems, AVs are capable of perceiving their environment, making real-time decisions, and executing driving tasks with remarkable precision. The impact of autonomous driving goes beyond convenience, with the potential to drastically improve road safety by minimizing the human errors that cause most traffic accidents. Furthermore, autonomous driving technologies promise to revolutionize transportation by enhancing traffic efficiency, reducing congestion, and lowering emissions through optimized driving and cooperative vehicle strategies. As these technologies continue to evolve, AVs are poised to redefine personal mobility, logistics, and public transportation, ushering in a future where vehicles act as intelligent, autonomous agents.

In this chapter, we will build upon the foundational concepts discussed earlier, focusing specifically on the methodologies and algorithms used in the components of AVs. While previous sections provided an overview of key components, this chapter will delve into how these elements are tailored and implemented within the context of autonomous driving. We will explore the specific perception technologies, planning and decision-making methodologies, and control algorithms that enable AVs to navigate complex environments safely and efficiently. Through this focused examination, readers will gain a deeper understanding of how current

technologies are applied to AVs, highlighting the state-of-the-art approaches that are shaping the future of transportation.

2.3.1 System Architecture in AVs

Autonomous driving systems rely on two fundamental design approaches: layered (modular) architecture and end-to-end learning. Both have distinct advantages and challenges, and they approach the problem of self-driving in different ways. Variations in architectural design result in differing data processing workflows and decision-making dynamics within AVs.

2.3.1.1 Modular (Layered) Architecture

Layered, or modular, architecture in autonomous systems organizes the system into distinct layers or modules, each responsible for a specific function, such as perception, decision-making, and control. This architecture promotes scalability and flexibility, allowing individual components to be developed, tested, and upgraded independently without affecting the overall system. In the context of autonomous driving, layered architecture enhances reliability by enabling seamless integration of various sensor inputs, data processing algorithms, and control mechanisms. Each module processes its respective task before passing data to the next layer, facilitating a structured and efficient flow of information. This modular approach not only simplifies system maintenance but also allows for easier adaptation to new technologies and environments. Modular architecture has common key components such as:

- Perception, which includes submodules such as sensing and processing raw sensor data (cameras, LiDAR, and radar), object detection and tracking, semantic segmentation, localization, and mapping.
- Planning, navigation, and decision-making, which includes submodules such as path (route) planning, behavior (maneuver) planning, and motion (trajectory) planning.
- Control system, actuation, and interaction with the environment, which includes submodules such as lateral control, longitudinal control, and actuators.

- Communication system, which includes intra-vehicle communication and V2X communication.

As an example, Waymo, one of the leaders in autonomous driving, uses a modular architecture. The perception system processes sensor data to detect the environment. This information is then passed to a prediction module that anticipates the movements of objects. Next, the planning module decides the vehicle's trajectory, and the control module implements the trajectory in real time. Figure 2.1 depicts the modular architecture of an AV system. Advantages of modular design in autonomous systems and intelligent agents include several key benefits:

- Flexibility: Each module within the system can be independently designed, tested, and optimized. This modularity allows for incremental improvements without disrupting the entire system, facilitating iterative development and refinement.
- Transparency: The modular approach provides engineers with full control and visibility over each component of the system. This enhances the ease of debugging and troubleshooting, as issues can be isolated and addressed within individual modules rather than the entire system.
- Reusability: Certain modules, such as those responsible for perception or control, can be reused across multiple projects. This reusability reduces development time and allows for adaptation of existing components to new tasks or domains, thereby enhancing efficiency and scalability.

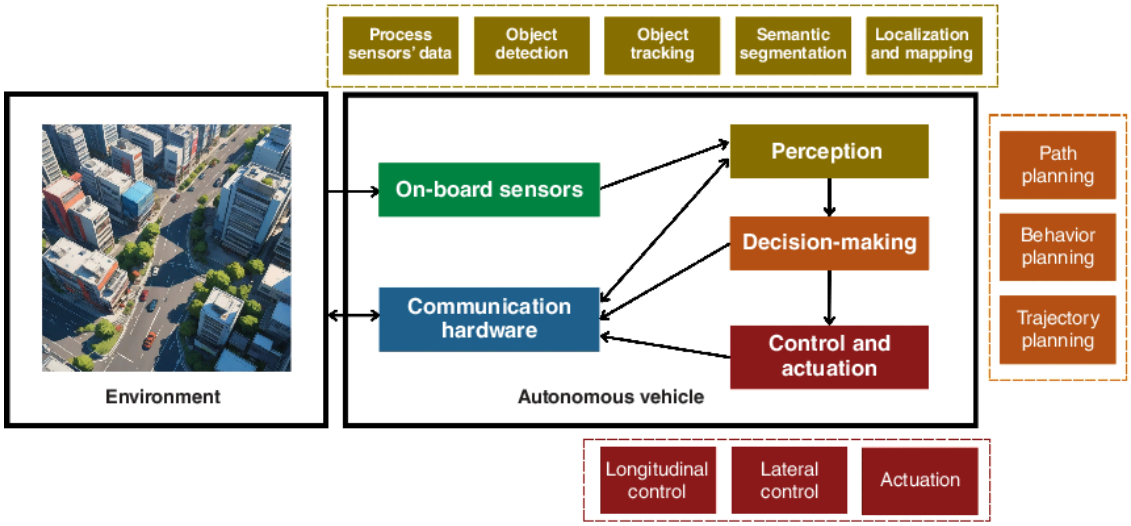


Figure 2.1 Modular architecture of an autonomous driving vehicle.

On the other hand, challenges and disadvantages of modular design in autonomous systems include several significant factors:

- Increased Complexity: The integration of multiple distinct modules necessitates careful system architecture and coordination. As more components are added, the overall system complexity grows, potentially leading to inefficiencies and difficulties in managing interactions between modules.
- Communication Overhead: The need for continuous data exchange between modules can introduce additional communication overhead. This inter-module communication may result in delays, particularly in real-time systems, or lead to a loss of precision as data is passed and processed sequentially across different components.

2.3.1.2 End-to-end Architecture

End-to-end architecture leverages deep learning to map raw sensor inputs, such as camera images, directly to vehicle actions such as steering, acceleration, and braking. Unlike traditional architectures that decompose tasks into distinct modules, a single neural network is trained to comprehend the entire task. A key component of this approach is a deep neural network that processes inputs and generates control actions. Adequate training data from diverse driving scenarios is essential, allowing the network to learn from human drivers' decisions in various conditions. Tesla's Autopilot exemplifies this approach, utilizing vast amounts of driving data to train deep neural networks capable of interpreting road

environments and making autonomous driving decisions without relying on predefined rules or modules. Hu et al. (2023) is among the state of the art of end-to-end architectures. There are several advantages for end-to-end architecture:

- Simplicity: The entire pipeline is replaced by a single model, which may simplify the system architecture.
- Learning from Data: End-to-end models can learn complex patterns and behaviors directly from data, allowing them to handle scenarios that modular systems may not anticipate.
- Improved Efficiency: With no need for communication between modules, end-to-end systems may be faster and less prone to communication errors.

Challenges of end-to-end architecture include, but are not limited to:

- Lack of Interpretability: It is harder to understand how the neural network is making decisions, which complicates debugging and troubleshooting.
- Data Requirements: End-to-end models require massive amounts of labeled training data to perform well across different driving environments and conditions.
- Safety Concerns: Since there's no explicit separation of concerns (e.g. perception, prediction), it's challenging to guarantee safety in all scenarios.

Hybrid Approaches: Some autonomous systems combine both approaches. For example, Tesla uses modular systems for certain tasks (e.g. perception) but relies on end-to-end deep learning for others (e.g. lane keeping and obstacle avoidance). This allows them to balance the strengths of each approach. In the following section, we delve into the components of modular architecture.

2.3.2 Perception in AVs

Perception in AVs is a multifaceted process that enables the vehicle to interpret and understand its environment. It begins with sensing, where data is collected from various sensors such as cameras, LiDAR, and radar.

This data is then processed for object detection, allowing the system to identify and classify surrounding objects such as vehicles, pedestrians, and obstacles. Once detected, multi-object tracking (MOT) follows, enabling the vehicle to monitor the movements of these objects over time to anticipate their future positions. Another essential process, semantic segmentation, divides the scene into distinct regions, helping the vehicle distinguish between road elements such as lanes, sidewalks, and curbs. Finally, localization and mapping ensure the vehicle accurately maps its environment and positions itself within it, using methods such as SLAM to navigate effectively. In the following sections, we will review each of these components.

2.3.2.1 Sensing

There are many sensors in AVs, such as cameras, LiDAR, radar, ultrasonic sensors, GPS, IMUs, and more to capture various types of information, such as visual data, depth, distance, speed, orientation, and environmental conditions. Autonomous driving systems, such as Waymo, rely heavily on three key sensor technologies: cameras, LiDAR, and radar, each with distinct advantages and limitations. Cameras excel at interpreting textures and colors, making them crucial for tasks such as lane detection, traffic sign recognition, and understanding the driving environment. However, processing high-resolution imagery demands significant computational resources. LiDAR sensors offer high precision, critical for object and pedestrian detection, as well as mapping for navigation and obstacle avoidance. Despite this precision, LiDAR faces challenges such as sensitivity to weather conditions, bulkiness, and high costs, although technological advances are gradually mitigating these issues. Radar, known for its robustness in adverse weather, provides reliable data on the distance and speed of nearby objects, making it essential for adaptive cruise control and collision avoidance. With a wide field of view and the ability to detect objects at long distances, radar complements other sensors but suffers from lower spatial resolution compared to LiDAR and reduced angular accuracy. Each of these technologies plays a vital role in ensuring the safety and effectiveness of AVs, balancing their strengths to compensate for individual weaknesses (Zhao et al. 2023).

2.3.2.2 Object Detection

The purpose of the object detection module is to identify and localize the surrounding road participants and traffic elements. In general, the object detection task can be categorized into two-dimensional (2D) and 3D methods. In 2D methods, the input is usually a RGB image obtained from a monocular camera. The main 2D object detectors used in the vehicular domain are variations of region-based CNNs, single-shot detectors, and transformer-based models.

- Region-based Convolutional Neural Networks (R-CNNs): R-CNNs are an early approach that use region proposals to identify potential objects, followed by classification through CNNs. Faster R-CNN improved detection speed by introducing shared convolutional layers and region proposal networks. This method provides high accuracy, which is crucial for safety in autonomous driving but is slower compared to newer methods (Girshick et al. 2014; Girshick 2015; Ren et al. 2016).
- Single-shot Object Detectors (YOLO and SSD): Single-shot object detectors, such as YOLO (You Only Look Once) (Redmon et al. 2016; Wang et al. 2024) and single-shot multiBox detector (SSD) (Liu et al. 2016), are known for their real-time performance. They treat object detection as a regression problem, predicting bounding boxes and class labels in a single stage. YOLO divides the image into grids and detects objects quickly, while SSD uses multiple feature maps to handle objects of various sizes. These methods are well-suited for real-time applications in AVs, where fast decision-making is critical.
- Transformer-based Object Detection (Detection Transformer [DETR]): Carion et al. (2020) introduced a transformer-based approach to object detection, eliminating the need for region proposals or anchor boxes. It processes the entire image in parallel, leveraging attention mechanisms to detect objects. While DETR simplifies the detection pipeline and shows promise in improving robustness, it is slower than YOLO and SSD, making it less suitable for real-time applications. Recently, real-time DETR was introduced by Zhao et al. (2024) to mitigate this problem.

3D object detection extends beyond 2D images, providing depth perception essential for understanding the spatial layout of the environment. It

relies on data from LiDAR sensors and stereo cameras, enabling AVs to detect objects in 3D space.

Point Cloud-based 3D Object Detection: Point cloud data, typically generated by LiDAR, represents the 3D environment. PointNet (Qi et al. 2017) and PointRCNN (Shi et al. 2019) are examples of models that directly process point clouds for object detection. PointNet operates on raw point clouds, while PointRCNN adds 3D region proposals to detect objects. These methods capture fine details in 3D space, which is crucial for detecting vehicles, pedestrians, and obstacles in autonomous driving.

Voxel-based 3D Object Detection: Voxelization divides the 3D space into a grid of voxels, and CNNs are applied to this structured data to detect objects. VoxelNet (Zhou and Tuzel 2018) utilizes this method to convert point cloud data into a voxel grid. This approach provides a structured representation of 3D space, making it effective for detecting objects at various distances and scales.

Both 2D and 3D object detection methods are integral to autonomous driving systems. While 2D detection provides fast and accurate object identification, 3D detection adds crucial depth perception, allowing vehicles to navigate complex environments safely. Together, these technologies drive the advancement of autonomous driving.

2.3.2.3 Multi-object Tracking

MOT is enabling the system to monitor and predict the movements of multiple objects, such as vehicles, pedestrians, and cyclists, within the vehicle's surroundings. By continuously tracking these objects, autonomous driving systems can anticipate potential collisions, adjust driving strategies, and navigate complex traffic scenarios. MOT typically follows object detection, where identified objects are assigned unique IDs and their movements are tracked over time. The system uses various algorithms, including Kalman filters, optical flow, and deep learning-based approaches, to estimate the trajectory of each object in real time. Kalman filters are widely used for predicting the future positions of objects based on their past movement, while deep learning models such as recurrent neural net-

works and Trackformer (Meinhardt et al. 2022) can handle more complex dynamics.

The challenges of MOT in AVs include dealing with occlusions, where objects may be temporarily hidden, and managing real-time performance, as delays in tracking can affect the vehicle's ability to make timely decisions. Reliable MOT systems must be robust in diverse conditions, such as high-speed driving or crowded urban environments, ensuring that the AV can operate safely and smoothly in dynamic, real-world situations.

2.3.2.4 Semantic Segmentation

Semantic segmentation is a crucial task in autonomous driving, enabling the vehicle to understand and interpret its environment by classifying each pixel in an image into predefined categories such as roads, vehicles, pedestrians, traffic signs, and more. This pixel-wise classification provides a detailed understanding of the scene, allowing the AV to differentiate between various objects and surfaces, which is essential for safe navigation.

Unlike traditional object detection, which only identifies and locates objects, semantic segmentation provides comprehensive context by labeling every part of the scene. This helps the vehicle to precisely recognize drivable areas, sidewalks, lane markings, and potential obstacles, enhancing decision-making in complex environments.

State-of-the-art deep learning models, such as fully convolutional networks (FCNs) (Long et al. 2015), U-Net (Ronneberger et al. 2015), DeepLab (Chen et al. 2017), and Mask R-CNN (He et al. 2017) and more recent architectures such as Segformer (Xie et al. 2021) and Oneformer (Jain et al. 2023) are widely used for semantic segmentation in autonomous driving. These models leverage advanced techniques like dilated convolutions and attention mechanisms to improve accuracy in identifying road elements and obstacles in real time.

Challenges in semantic segmentation include dealing with varying lighting conditions, weather changes, and occlusions in real-world scenarios. Despite these challenges, it plays a vital role in improving the perception

and planning capabilities of AVs, contributing to their ability to safely navigate diverse and dynamic environments.

2.3.2.5 Localization and Mapping

Localization and mapping enable AVs to understand and update their precise location in their environment and to build a map for navigation. Localization involves determining the vehicle's exact position on the road, while mapping creates a detailed representation of the surroundings, including roads, obstacles, and landmarks. These processes work together to ensure the vehicle can safely navigate through dynamic environments.

State-of-the-art techniques for localization and mapping rely on sensors such as GPS, LiDAR, cameras, and IMUs. One prominent method is SLAM (Thrun 2008), which allows the vehicle to build and update a map while simultaneously localizing itself within it. SLAM algorithms, such as graph-based SLAM and particle filter SLAM, are widely used for real-time applications in autonomous driving. Visual SLAM (using cameras) and LiDAR-based SLAM are leading technologies, offering high precision in both indoor and outdoor environments. Advanced deep learning-based approaches, such as DeepVo (Wang et al. 2017) and VoxelNet (Zhou and Tuzel 2018), further enhance localization by integrating vision and 3D point cloud data.

Challenges include dealing with sensor noise, GPS outages, and dynamic obstacles, but state-of-the-art SLAM and deep learning models are continually improving the accuracy and robustness of localization and mapping, driving the evolution of AVs.

2.3.3 Planning and Decision-making

Planning and decision-making are essential components of autonomous driving, enabling vehicles to navigate complex environments safely and efficiently. These processes are divided into three key stages: path planning, behavior planning, and motion planning. Each stage addresses dif-

ferent aspects of driving, from determining the overall route to executing fine-grained maneuvers.

2.3.3.1 Path (Route) Planning

Path planning involves generating a long-term route from the vehicle's current location to the destination, typically using GPS data and high-definition (HD) maps. This stage accounts for road networks, traffic conditions, and environmental factors such as road closures or accidents. Path planning algorithms, such as Dijkstra's algorithm and A*, are commonly used to compute the shortest and most efficient route based on road geometry and traffic data. State-of-the-art methods include the use of AI and deep learning to enhance traditional algorithms. For example, Google's Waymo and Tesla's Autopilot systems incorporate predictive models that forecast traffic flow and road conditions, allowing the vehicle to make dynamic adjustments to the route. Additionally, the use of HD maps enables vehicles to understand detailed road structures, including lane markings, speed limits, and intersections, allowing more precise route planning. The integration of real-time traffic data and predictive modeling significantly improves the vehicle's ability to select the safest and most efficient path.

2.3.3.2 Behavior Planning

Behavior planning is responsible for making high-level decisions on how the vehicle interacts with its surroundings. This includes lane changes, overtaking, merging, and reacting to pedestrians or obstacles. Behavior planning must balance safety, efficiency, and comfort by determining the appropriate action in response to dynamic traffic scenarios.

State-of-the-art behavior planning systems use ML and rule-based approaches. For instance, Waymo and Uber's autonomous driving systems employ RL to simulate various driving scenarios, enabling the vehicle to learn optimal decision-making strategies. Another key method is the use of partially observable Markov decision processes, which help the vehicle account for uncertainty in the environment, such as predicting the behavior of other drivers or pedestrians. These models allow the system to

make proactive and reactive decisions in real time, improving the vehicle's ability to handle complex driving tasks.

2.3.3.3 Motion (Trajectory) Planning

Motion planning, or trajectory planning, generates detailed paths that the vehicle follows to execute the behavior plan. This stage takes into account the vehicle's kinematic constraints, road geometry, and the surrounding environment to ensure safe, smooth, and efficient movement. The motion planner must also ensure that the trajectory is feasible for the vehicle's dynamics while providing a comfortable ride for passengers. State-of-the-art motion planning algorithms include MPC and rapidly-exploring random trees (RRTs). MPC is a popular method used in AVs such as Tesla's Autopilot, where the algorithm continuously optimizes the vehicle's trajectory based on real-time sensor inputs. RRT, on the other hand, is particularly useful for complex environments, as it explores potential paths and refines them into smooth trajectories. Tesla's Full Self-Driving (FSD) system utilizes a hybrid approach, combining both methods to generate collision-free, comfortable, and safe trajectories. Moreover, advancements in deep RL have been applied to motion planning to enhance the system's ability to adapt to rapidly changing environments. This allows the vehicle to dynamically adjust its trajectory when unexpected obstacles or traffic patterns emerge.

In summary, planning and decision-making process in autonomous driving is a multilayered approach that ensures safe and efficient navigation. Path planning provides a global route, behavior planning makes high-level decisions, and motion planning generates precise trajectories. State-of-the-art methods, such as RL, MPC, and hybrid models, are continuously advancing the capabilities of AVs. These technologies enable autonomous systems to navigate complex environments while ensuring safety, comfort, and efficiency, moving closer to fully autonomous driving scenarios (see Grigorescu et al. 2020; Teng et al. 2023).

2.3.4 Control System, Actuation, and Interaction with the Environment

The control system in autonomous driving is critical for executing the planned trajectory by controlling the vehicle's movements and interactions with its surroundings. This involves managing steering, acceleration, and braking, ensuring that the vehicle follows the desired path while maintaining safety, comfort, and efficiency. Control systems are divided into lateral and longitudinal controls, both of which interact with actuators that carry out the physical commands. While traditional methods such as MPC are commonly used, RL is gaining prominence for more adaptive and dynamic control strategies.

2.3.4.1 Lateral Control

Lateral control manages the steering to keep the vehicle on the correct trajectory. This includes tasks such as lane keeping, cornering, and responding to complex road conditions. MPC is widely used for lateral control due to its ability to predict the vehicle's future state and adjust the steering accordingly. MPC ensures smooth, precise steering adjustments, which are essential for lane following and handling turns.

However, RL is increasingly being explored for lateral control in more dynamic environments. Deep reinforcement learning (DRL) techniques, such as deep Q-networks (DQN) and proximal policy optimization (PPO), allow the vehicle to learn optimal steering strategies through trial and error in simulation environments. For example, RL can help the vehicle adapt its steering to handle sharp curves, avoid obstacles, or navigate through complex traffic patterns more flexibly than traditional methods. Some research shows that RL-based lateral control can outperform MPC in situations requiring rapid, adaptive decision-making, though safety and real-time performance remain challenges.

2.3.4.2 Longitudinal Control

Longitudinal control governs the vehicle's speed through acceleration and braking, ensuring that the vehicle maintains a safe distance from other road users while following the planned trajectory. Traditional methods

such as adaptive cruise control (ACC) and PID controllers are commonly used. ACC adjusts the vehicle's speed based on the distance to the vehicle ahead, and PID controllers provide precise adjustments to speed and braking, ensuring smooth transitions and passenger comfort.

In addition to these traditional methods, RL is also being applied to longitudinal control. RL can optimize the balance between safety, comfort, and energy efficiency by learning from interactions with the environment. For instance, deep Q-learning can help the vehicle decide when to accelerate or brake in complex traffic, improving energy efficiency or reducing sudden braking. RL can also adapt to unusual driving scenarios, such as aggressive or erratic driving from other vehicles, where traditional controllers may struggle to respond quickly and efficiently.

2.3.4.3 Actuators

Actuators are the physical components that execute the control commands from the vehicle's software, including steering, throttle, and braking systems. The steering actuator is typically controlled by electric power-assisted steering systems, allowing precise and responsive control based on lateral control inputs. The throttle and braking actuators are responsible for regulating speed and stopping the vehicle, usually through electronic throttle control and electronic brake control systems.

State-of-the-art methods incorporate RL to optimize actuator control for better performance under varying conditions. For instance, RL can fine-tune how actuators respond to dynamic situations such as slippery roads or sudden obstacles. Autonomous driving systems, such as those in Tesla's FSD suite, use a combination of traditional methods and RL to control actuators, ensuring a smooth balance between safety, responsiveness, and energy efficiency. The control system in autonomous driving plays a pivotal role in translating high-level planning and decision-making into real-world actions. While traditional methods such as MPC and PID control are still dominant in lateral and longitudinal control due to their reliability and predictability, RL is gaining ground for more dynamic, adaptive control. RL methods such as deep Q-learning and proximal policy optimization (PPO) (Schulman et al. 2017) are particularly useful in complex

or unpredictable environments, where learning from interactions with the environment can lead to better performance. Advances in actuators, combined with RL, further enhance the vehicle's ability to adapt to varying road conditions, contributing to more efficient and safer autonomous driving.

2.3.5 Vehicular Communication and Cooperative Intelligence

Effective vehicular communication systems are essential for autonomous driving, enabling the exchange of crucial data within the vehicle and with the external environment. These systems ensure real-time decision-making, coordination, and safety in complex driving scenarios.

V2X Communication: V2X communication enables the vehicle to communicate with its external environment, including other vehicles (V2V), infrastructure (V2I), and even pedestrians (V2P). V2X communication is critical for improving safety and enabling cooperative driving in high-traffic or complex scenarios.

State-of-the-art V2X technologies include dedicated short-range communications (DSRC) and cellular V2X (C-V2X). DSRC is a mature technology designed for low-latency communication, making it ideal for safety-critical applications like collision avoidance. C-V2X, on the other hand, leverages 5G cellular networks to provide high-bandwidth, long-range communication, allowing vehicles to communicate with infrastructure for traffic management or with pedestrians via smartphones.

These advanced communication systems enable cooperative intelligence, where AVs coordinate their actions, such as optimizing traffic flow, reducing accidents, and enabling cooperative maneuvers such as platooning.

2.3.5.1 Cooperative AI in Autonomous Driving

Cooperative AI in autonomous driving is a rapidly evolving field that improves the safety, efficiency, and comfort of AVs by going beyond the autonomy of individual vehicles. It enables vehicles to collaborate, share information, and make coordinated decisions. This system relies on communication between V2V, V2I, and V2X, creating a more connected and

efficient transportation network. Two key components of cooperative AI are cooperative perception and cooperative planning, both of which depend on reliable communication technologies.

Cooperative perception enables AVs to enhance their sensory capabilities by sharing data with nearby vehicles, infrastructure, and pedestrians or cyclists with smart devices. While traditional perception relies on onboard sensors such as LiDAR, cameras, radar, and ultrasonic sensors, these technologies are limited by range, line of sight, and environmental conditions such as fog or rain. By adopting cooperative perception, vehicles can mitigate these limitations through real-time data sharing. For instance, if a vehicle detects a hazard, such as an icy patch or a pedestrian hidden from view, it can quickly communicate this information to trailing vehicles. This allows them to anticipate and respond to hazards much sooner than their own sensors would have detected them.

- Early research on cooperative perception explored data sharing at two levels: early-stage sharing (Arnold et al. 2020; Chen et al. 2019b), where raw sensory data such as LiDAR point clouds were exchanged, and late-stage sharing, where locally processed information (Günther et al. 2016; Hobert 2015), such as object detection outputs, was shared among agents. While late-stage data sharing requires minimal communication bandwidth, it is less effective in scenarios involving partial occlusion, where a hazard may not be visible to all agents. On the other hand, early-stage sharing, although more robust against occlusion, demands significantly higher communication bandwidth, making it impractical in many real-world applications. To address these challenges, state-of-the-art cooperative perception methods have found a balance by developing fusion algorithms that allow the sharing and integration of intermediate features from the neural networks used in perception tasks. This approach strikes a compromise, reducing bandwidth requirements while still improving the system's ability to handle occlusions effectively (Marvasti et al. 2020; Chen et al. 2019a; Wang et al. 2020a; Xu et al. 2022; Wang et al. 2023). Figure 2.2 demonstrates different stages of information sharing for cooperative perception task.

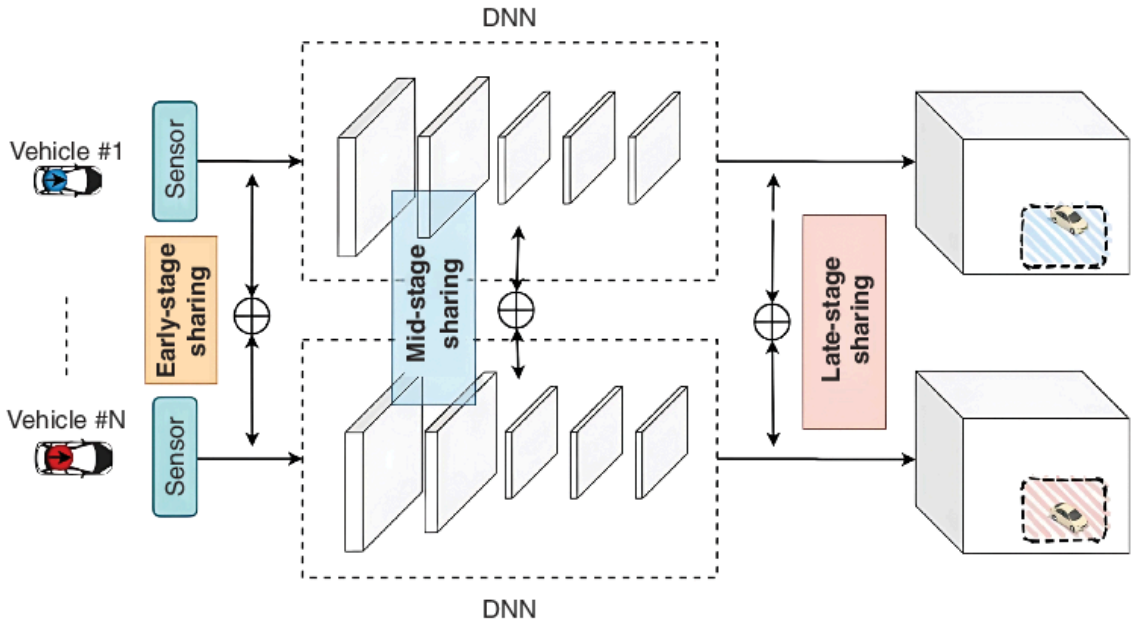


Figure 2.2 Overview of cooperative perception by sharing information at different stages. In early-stage sharing methods, the raw sensor data is shared between agents. This data is aligned and aggregated at the receiver for perception. In mid-stage sharing methods, the raw information is fed to a neural network, and the extracted features are shared amongst agents. In late-stage sharing, each agent runs perception algorithms based on its own onboard sensory data and shares the output with others.

Source: Adapted from Marvasti et al., 2020 /arXiv).

Cooperative planning extends coordination beyond individual vehicle decisions, addressing the limitations of traditional autonomous driving, where each vehicle independently plans its route, speed, and maneuvers. This isolated approach often leads to inefficiencies or conflicts, especially in high-traffic scenarios. In cooperative planning, vehicles communicate their intentions, e.g. lane changes, turns, or speed adjustments, and collaborate to achieve shared goals. For example, when approaching an intersection, vehicles can coordinate their paths and speeds to cross without stopping or dynamically negotiate right of way.

This coordinated approach provides several important benefits. First, by sharing planned actions such as lane changes or stops, vehicles can adjust their trajectories in real time, enhancing collision avoidance and creating a proactive safety mechanism (Chen et al. 2024). Additionally, traffic flow optimization is achieved through smoother merges, exits, and lane changes, reducing congestion and minimizing stop-and-go driving (Toghi et al. 2021; Valiente et al. 2022). On highways, vehicles can form tightly packed platoons, synchronizing movements to lower air drag, improve fuel efficiency, and increase road capacity (Duan et al. 2023; Razzaghpour et al. 2023). Furthermore, adaptive traffic management allows vehicles to

dynamically reroute and adjust speeds based on real-time traffic data, reducing travel time and energy consumption, benefiting both individual vehicles and the overall network (Guo et al. 2023; Wang et al. 2020b).

2.4 Challenges and Future Directions

Autonomous systems and intelligent agents face significant challenges that need to be addressed for widespread adoption. One of the primary challenges is ensuring safety and reliability in complex, dynamic environments. AVs, for example, must operate seamlessly in diverse conditions, such as inclement weather, heavy traffic, and unpredictable pedestrian behavior. Ensuring that these systems can handle rare, edge-case scenarios remains a critical hurdle.

Another challenge is data privacy and security. As autonomous systems rely heavily on data from various sources, including sensors, infrastructure, and personal devices, ensuring that this data is protected from cyberattacks is essential. Robust cybersecurity frameworks are needed to prevent hacking or malicious attacks that could compromise system performance.

Regulatory and legal frameworks are also lagging behind technological advancements. Defining responsibility in cases of accidents or system failures is still an unresolved issue, requiring collaboration between governments, industries, and policymakers.

In terms of future directions, ML and AI improvements will be pivotal in enhancing the decision-making abilities of intelligent agents. Multimodal large language models (Cui et al. 2024) are another promising area, allowing autonomous systems to provide human-level perception, reasoning, and decision-making capabilities. Lastly, advancements in cooperative intelligence and V2X communication will enable more coordinated actions between autonomous systems, leading to more efficient and safer operations.

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